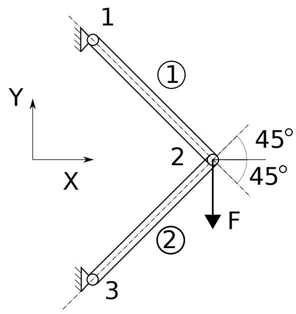
LUT University Summer 2026
BK10A6400 Basics of FE Analysis
Marko Matikainen

Bonus task 3 1.6.2026
Local and global stiffness matrices, boundary conditions, and loadings.
Possibility to earn five bonus points
Deadline: Before the lectures on the 4th of June

The aim is to solve global nodal displacements u_2, v_2 at node 2 of a simple truss structure shown in the Figure, and after solving nodal displacements, solve member forces (elemental level) for rods 1 and 2. Use numbering for nodes and elements that are shown in the Figure.



The parameters of the problem are given in the Table.

	Young's modulus, E [MPa]	Cross section, A [mm^2]	Length, L [mm]
Truss 1	210000	100	1000
Truss 2	210000	50	1000
Force, F [N]	100 000		

Write a short report where you give answers for the following:

1. How the element's angles is defined (e.g., measured counter-clockwise from the global x-axis) and illustrate this with a figure showing the geometry of structure, node and element numbering, coordinate definitions and angles (1 point).
2. Determinant of the reduced global stiffness matrix after enforcing boundary conditions by elimination (1 point).
3. Global nodal displacements (u, v) at node 2 (1 point).
4. Local elemental nodal displacements for each rod (1 point total):
   - Rod 1: $(\overline{u}_1^{(1)}, \overline{u}_2^{(1)})$
   - Rod 2: $(\overline{u}_1^{(2)}, \overline{u}_2^{(2)})$
5. Member (axial) forces $\overline{\boldsymbol{f}}^{(i)}$ for rods $i \in \{1, 2\}$ (1 point total). Clearly state whether each rod is in **tension** or **compression**, based on the sign of the computed elemental forces.

Present all numerical results in tables (include units where applicable) and submit in Moodle:

- A PDF report.
- All MATLAB (`.m`) or Python (`.py`) source files required to reproduce your results.

Make sure that the code runs without extra editing (provide a main script and functions).

Note: No points will be awarded without both the PDF report and runnable source code.

Please note that code templates KlocRod.m, TRod.m, and run_linStaticRod.m for MATLAB and KlocRod.py, TRod.py, and run_linStaticRod.py for Python are provided.

Tips

In this task, you can use a ready solution for a stiffness matrix. As you know from lectures and literature (see for example, Cook p.20 Eq. (2.2.1) or Hakala p.54 Eq. (2.3.17)), the stiffness matrix of a rod element can be written as follows:

$$\overline{\boldsymbol{K}} = \frac{EA}{L}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \tag{1}$$

Analyzed structures are not often parallel to the element's longitudinal direction -> local elemental stiffnesses need to be present in the global coordinate system.

$$\boldsymbol{K}_G = \boldsymbol{T}^T\overline{\boldsymbol{K}}\boldsymbol{T} \tag{2}$$

where $\boldsymbol{K}_G$ is a global elemental stiffness matrix (of size 4×4 for a two-nodded rod element) and $\boldsymbol{T}$ is a transformation matrix that can be written as follows:

$$\boldsymbol{T} = \begin{pmatrix} \cos\alpha & \sin\alpha & 0 & 0 \\ 0 & 0 & \cos\alpha & \sin\alpha \end{pmatrix} \tag{3}$$

Transformation needs to be done for all (two) elements.

$$\boldsymbol{K}_G^{(1)} = \boldsymbol{T}^T(\alpha_1)\overline{\boldsymbol{K}}^{(1)}(E_1, A_1, L_1)\boldsymbol{T}(\alpha_1) \tag{4}$$

$$\boldsymbol{K}_G^{(2)} = \boldsymbol{T}^T(\alpha_2)\overline{\boldsymbol{K}}^{(2)}(E_1, A_1, L_1)\boldsymbol{T}(\alpha_2) \tag{5}$$

and global elemental stiffness matrices look like that

$$\boldsymbol{K}_G^{(1)} = \begin{pmatrix} k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} \\ k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} \\ k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} & k_{34}^{(1)} \\ k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} & k_{44}^{(1)} \end{pmatrix} \tag{6}$$

$$\boldsymbol{K}_G^{(2)} = \begin{pmatrix} k_{11}^{(2)} & k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} \\ k_{21}^{(2)} & k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} \\ k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} \\ k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} \end{pmatrix} \tag{7}$$

Assembling

Sometimes, it is helpful to present elements' connectivity and corresponding (global) nodal displacements in the table as follows:

	Node I	Node J	nodal displacements
Truss 1	1	2	u_1, v_1, u_2, v_2
Truss 2	2	3	u_2, v_2, u_3, v_3

Note that $u_1, v_1, u_2, v_2, u_3, v_3$ are the structure's global displacement coordinates. Common stiffness components (interconnected via the same node) of global elemental stiffness matrices $\boldsymbol{K}^{(1)}$ and $\boldsymbol{K}^{(2)}$ can be seen from the connectivity table and they are needed to sum to mimic structure correctly. After assembling, the global stiffness matrix should be a size of 6×6.

Boundary conditions and linear solution

The rods are fixed to the wall via hinged supports (at nodes 1 and 3). Therefore, global nodal displacements $u_1 = 0, v_1 = 0, u_3 = 0, v_3 = 0$ are restricted to displace. After eliminating corresponding rows and columns of the global stiffness matrix and external force vector, global displacements can be solved:

$$\boldsymbol{u}_{G_{red}} = \boldsymbol{K}_{G_{red}}^{-1} \boldsymbol{f}_{G_{red}} \tag{8}$$

Member forces

After solving $\boldsymbol{u}$, local elemental displacements for an element i can be compute as follows:

$$\overline{\boldsymbol{u}}^{(i)} = \boldsymbol{T} \boldsymbol{u}_G^{(i)} \tag{9}$$

and based on this, member forces for an element i can be found as

$$\overline{\boldsymbol{f}}^{(i)} = \overline{\boldsymbol{K}}^{(i)} \overline{\boldsymbol{u}}^{(i)} \tag{10}$$

So, local elemental level displacements for the rod 1:

$$\overline{\boldsymbol{u}}^{(1)} = \boldsymbol{T}(\alpha_1) \boldsymbol{u}^{(1)} \tag{11}$$

Note that $\boldsymbol{u}^{(1)}$ should include all global nodal displacements of the rod (i.e element in this case) 1 (solved nodal coordinates and nodal coordinates at supports). Member forces for the rod 1 can be found as:

$$\overline{\boldsymbol{f}}^{(1)} = \overline{\boldsymbol{K}}^{(1)} \overline{\boldsymbol{u}}^{(1)} \tag{12}$$

and same operations for a rod 2...